

Smart Sunburn Warning Patch: A Wearable Multi-Sensor System for Real-Time UV Exposure Monitoring and Risk Alerting

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Abstract—Excessive ultraviolet (UV) radiation exposure is a leading cause of sunburn, premature skin ageing, and melanoma, yet no compact, body-worn monitoring device currently exists that provides personalised, real-time warnings. This paper presents the *Smart Sunburn Warning Patch*, a wearable embedded system that fuses data from a UV sensor (ML8511), a precision temperature sensor (TMP117), and an ambient light sensor (BH1750) to compute a continuous skin-exposure risk score. An ESP32-S3 microcontroller performs on-board sensor fusion and drives a three-level LED alert system (green / amber / red) together with a vibration motor for haptic feedback. The PCB measures 39 mm × 28 mm, operates from a single 3.3 V regulated rail derived from a 3.7 V Li-Po battery, and incorporates a mixed-signal layout strategy to minimise noise coupling. The design demonstrates that low-cost, commercially available components can be integrated into a clinically relevant wearable that addresses the limitations of coarse, location-based UV index applications. The complete hardware design files and firmware source code are available at <https://github.com/Sidd-Devs/Smart-Sunburn-Warning-Patch>.

Index Terms—UV monitoring, wearable electronics, embedded systems, sensor fusion, ESP32-S3, sunburn prevention, PCB design.

I. INTRODUCTION

Ultraviolet radiation in the UVA (315–400 nm) and UVB (280–315 nm) bands is responsible for a wide spectrum of dermatological harm. Short-term overexposure causes erythema (sunburn), while chronic exposure is strongly correlated with squamous cell carcinoma and malignant melanoma—cancers whose global incidence has risen steadily over recent decades [1]. Despite widespread public awareness, individuals routinely underestimate their UV dose because UV is invisible and environmental factors—cloud cover, reflective surfaces such as water and sand, and altitude—can produce misleading conditions where perceived brightness gives no indication of actual UV intensity [2].

Existing consumer-facing solutions, principally smartphone applications that report the UV index (UVI) forecast for a given geographic location, suffer from three fundamental limitations. First, they report a spatially averaged, forecast quantity rather than a measurement at the body surface. Second, they cannot account for individual skin phototype, clothing coverage, or localised exposure geometry. Third, they rely on the user

to actively consult the application rather than providing a proactive, ambient warning [3].

Wearable UV sensors have been explored in academic literature, yet commercially available implementations remain absent. This work addresses the gap by presenting a complete hardware design for a wearable patch that: (a) measures UV irradiance directly at skin level using the ML8511 analog UV sensor [4], (b) incorporates surface temperature and ambient illuminance as contextual inputs, (c) computes a fused risk score on-device using an ESP32-S3 microcontroller [5], and (d) alerts the wearer through a graded visual and haptic interface—all within a PCB footprint suitable for adhesive patch deployment.

The remainder of this paper is organised as follows. Section II describes the system architecture, component selection, hardware design choices, and PCB implementation. Section III presents the functional verification and performance characterisation. Section IV concludes and identifies directions for future work.

II. SYSTEM DESIGN

A. System Architecture Overview

The Smart Sunburn Warning Patch follows a sense–convert–process–alert pipeline, illustrated in Fig. 1. Three sensors feed the ESP32-S3 microcontroller: the ML8511 UV sensor via the on-chip 12-bit ADC, and the TMP117 temperature sensor together with the BH1750 ambient light sensor via a shared I²C bus. The microcontroller evaluates a weighted risk algorithm and drives alert actuators—an RGB LED and a vibration motor—through GPIO-controlled transistor drivers. All active circuitry is powered from a single regulated 3.3 V rail derived from a 3.7 V lithium-polymer cell via a low-dropout (LDO) regulator.

B. Sensor Selection and Justification

1) *ML8511 UV Sensor*: The ML8511 (ROHM Semiconductor) converts incident UV irradiance in the 280–390 nm band (covering both UVA and UVB) directly to an analog output voltage proportional to UV intensity, spanning 0–3.3 V across the ADC input range of the ESP32-S3 [4]. Its low supply current, compact SOT-23-5 package, and direct ADC

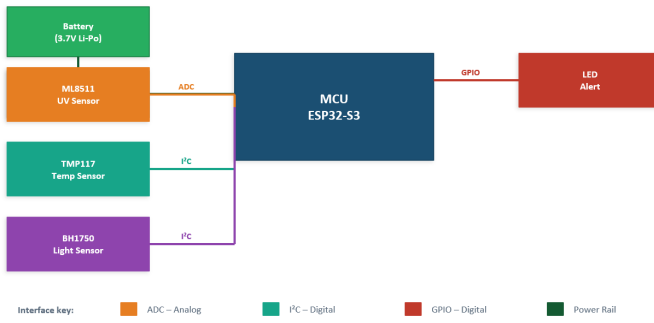


Fig. 1. System architecture block diagram showing power rail, analog ADC path (ML8511), I²C digital bus (TMP117, BH1750), and GPIO alert outputs from the ESP32-S3.

compatibility make it well-suited for wearable integration without additional signal-conditioning circuitry. As the primary driver of sunburn, UV irradiance is the highest-weighted input to the risk model.

Fig. 2 shows the ML8511 sub-schematic. The EN (enable) pin is tied to VDD to keep the sensor in continuous measurement mode. The analog output (OUT, pin 8) is routed directly to the ESP32-S3 ADC net. Two decoupling capacitors—C7 (100 nF ceramic) and C8 (1 nF ceramic)—are placed in parallel on the VDD supply pin to suppress high-frequency transients and low-frequency supply ripple respectively, ensuring a stable reference for the analog output. Pins 1–3, 6, 10–12 are no-connect (NC), consistent with the manufacturer’s recommended application circuit.

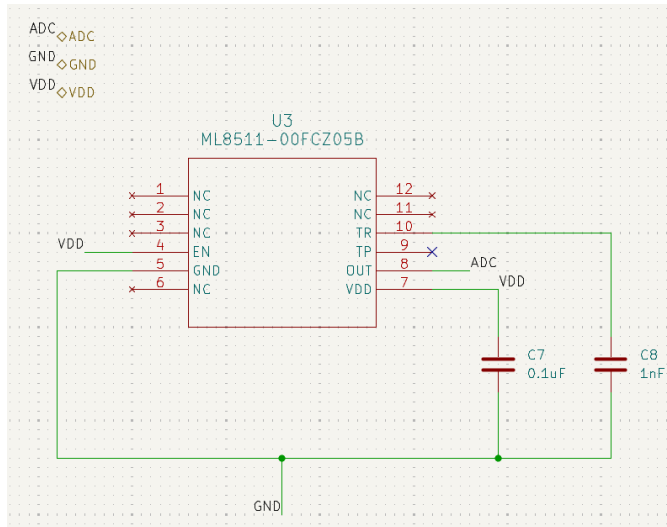


Fig. 2. ML8511 UV sensor sub-schematic (U3). The OUT pin routes to the ESP32-S3 ADC. Decoupling is provided by C7 (100 nF) and C8 (1 nF) on the VDD pin. The EN pin is pulled to VDD for continuous operation.

2) *TMP117 High-Accuracy Temperature Sensor*: The TMP117 (Texas Instruments) provides temperature measurements with $\pm 0.1^\circ\text{C}$ accuracy over its full operating range and a resolution of 0.0078°C per least-significant bit of its 16-bit output register. Communication occurs over I²C at up

to 400 kHz. Skin surface temperature serves as a proxy for cumulative thermal exposure and modulates the severity of UV-induced damage; incorporating temperature into the algorithm therefore improves the fidelity of the risk prediction beyond what UV irradiance alone can provide [2].

Fig. 3 shows the TMP117 sub-schematic. The device (U2, TMP117xxDRV) is powered from the 3.3 V VDD rail with the ADDR address pin tied to GND, assigning the default I²C address (0x48). The ALERT output pin (pin 3) is pulled up to VDD through a $5\text{ k}\Omega$ resistor (R12) and exported as the ALERT net to notify the MCU of threshold-crossing events independently of the I²C polling cycle. The SDA and SCL lines connect to the shared I²C bus along with the BH1750.

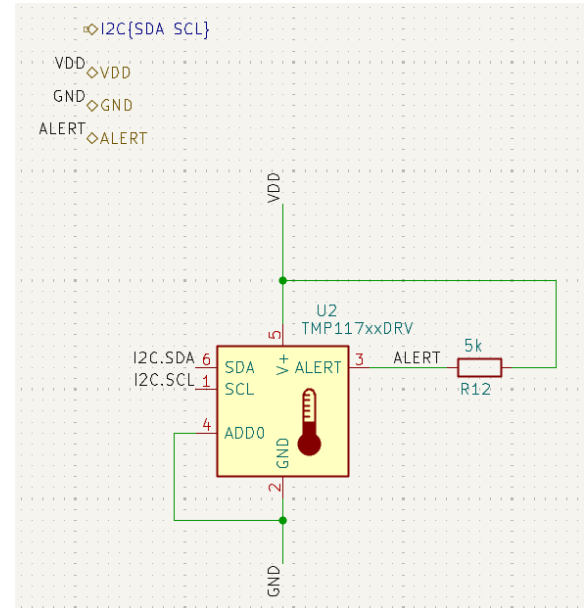


Fig. 3. TMP117 temperature sensor sub-schematic (U2). The open-drain ALERT pin is pulled up via R12 ($5\text{ k}\Omega$) and exported to the MCU for interrupt-driven threshold detection.

3) *BH1750 Ambient Light Sensor*: The BH1750FVI-TR (ROHM Semiconductor) is a 16-bit digital light sensor with an operating range of 1–65 535 lux and an I²C interface operating at 400 kHz. It requires no factory calibration. Ambient illuminance provides environmental context that allows the firmware to distinguish outdoor conditions (high lux, meaningful UV) from indoor or overcast conditions (low lux, suppressed UV), thereby reducing the rate of false alerts and improving overall system specificity [3].

Fig. 4 shows the BH1750 sub-schematic. The device (U5, BH1750FVI-TR) is powered from VDD with the ADDR selection pin (pin 3) tied to GND, assigning I²C address 0x23. A 100 nF decoupling capacitor (C10) is placed at the VCC pin, and an additional 100 nF capacitor (C9) is placed close to the I²C interface pins. A $1\text{ k}\Omega$ pull-up resistor (R6) is provided on the SCL line within the sub-schematic, complementing the global $4.7\text{ k}\Omega$ pull-ups on the shared bus. The SDA and SCL pins connect to the common I²C net shared with the TMP117.

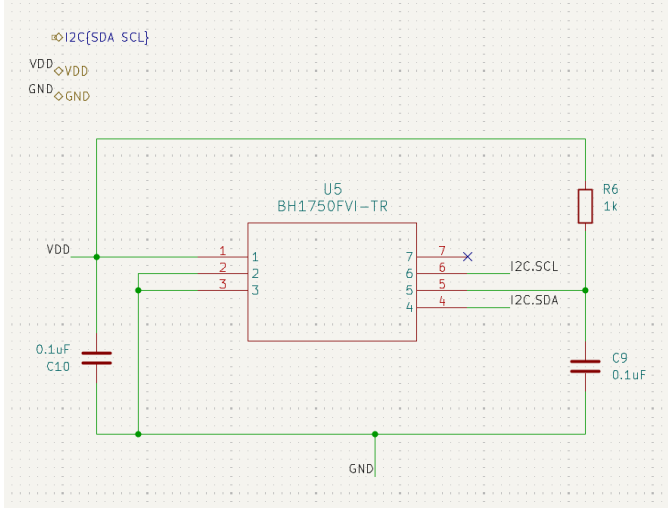


Fig. 4. BH1750FVI ambient light sensor sub-schematic (U5). R6 (1 kΩ) provides local SCL pull-up. C10 and C9 (both 100 nF) decouple the supply and I²C interface respectively.

TABLE I
SENSOR SPECIFICATIONS SUMMARY

Parameter	ML8511	TMP117	BH1750
Interface	ADC (analog)	I ² C	I ² C
Range / Accuracy	0–3.3 V out	±0.1 °C	1–65535 lux
Resolution	12-bit (ADC)	0.0078 °C	16-bit
Bus speed	–	400 kHz	400 kHz
I ² C address	–	0x48	0x23
Calibration required	No	No	No
Decoupling	100 nF + 1 nF	Via bus pull-ups	100 nF × 2

C. Microcontroller: ESP32-S3

The ESP32-S3 (Espressif Systems) was selected as the central processing unit. Key attributes relevant to this design include a dual-core Xtensa LX7 processor operating at up to 240 MHz, an integrated 12-bit successive approximation register (SAR) ADC for the ML8511 analog output, native I²C peripheral support, integrated Wi-Fi and Bluetooth Low Energy (BLE) for future wireless connectivity, and 3.3 V I/O compatibility that eliminates the need for voltage-level translation between the MCU and peripheral sensors [5].

Fig. 5 shows the ESP32-S3-WROOM-1-N16 module sub-schematic (U1). The design includes a push-button reset circuit (SW1) on the EN line with a 1 kΩ series resistor (R16) and a 22 µF decoupling capacitor (C1) to ensure clean power-on-reset behaviour. Two additional push buttons (SW6, SW2) are connected to GPIO pins for boot-mode selection and user interaction. The I²C bus (I2C.SDA on IO12, I2C.SCL on IO17) is pulled up through 2.2 kΩ resistors (R1, R8) on the MCU sheet, supplementing the local pull-ups on each sensor sub-schematic. GPIO0 drives an onboard LED (D6) through a 1 kΩ current-limiting resistor (R11). Supply decoupling is provided by C32 (22 µF), C31 (100 nF), and C33 (22 µF)

distributed across the module VDD pins, while C28 and C29 (both 100 nF) decouple the USB D+ and D– interface lines. A 1 kΩ resistor (R18) on the 3V3 rail provides inrush current limiting from the LDO output.

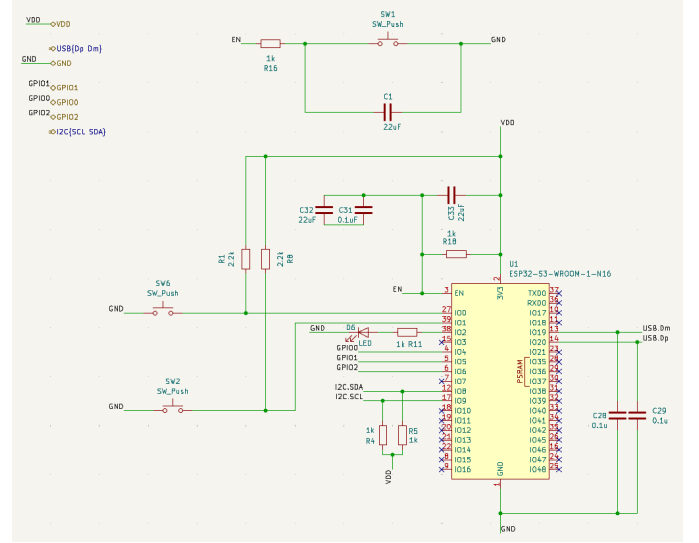


Fig. 5. ESP32-S3-WROOM MCU sub-schematic (U1). Key features: reset circuit (SW1, R16, C1); I²C pull-ups (R1, R8 – 2.2 kΩ) on IO12/IO17; onboard LED (D6, R11) on GPIO0; multi-stage VDD decoupling (C32, C31, C33); and USB interface decoupling (C28, C29).

D. Risk Scoring Algorithm

The firmware implements a weighted multi-parameter risk score defined as:

$$R = w_{UV} \cdot \hat{u} + w_T \cdot \hat{t} + w_L \cdot \hat{l} \quad (1)$$

where $\hat{u}$, $\hat{t}$, and $\hat{l}$ are normalised UV index, temperature deviation from a 25 °C baseline, and ambient lux values respectively, and w_{UV} , w_T , w_L are empirically tuned weighting coefficients. The UV term receives the highest weight since UV irradiance is the primary erythral driver [2]. The alert thresholds map onto three risk levels as summarised in Table II.

TABLE II
RISK LEVEL THRESHOLDS AND ALERT OUTPUTS

Risk Level	UV Index	Skin Temp.	Alert
Low	< 3	Normal	Green LED
Moderate	3–6	Elevated	Amber LED
High	> 6	High	Red LED + Vibration

E. Key Hardware Design Choices

Single 3.3 V Power Rail. All ICs operate at 3.3 V, supplied by an LDO regulator from the 3.7 V Li-Po cell. A unified rail eliminates level-shifting circuitry, simplifies the power distribution network, and reduces BOM complexity.

Mixed-Signal PCB Partitioning. The analog front-end section—comprising the ML8511, its ADC routing traces,

and associated decoupling—is physically separated from the I²C digital region. This partitioning minimises capacitive and inductive coupling between the switching-noise-prone digital domain and the sensitive analog signal path, preserving ADC accuracy [6].

I²C Bus Topology. The TMP117 and BH1750 share a single SDA/SCL bus, reducing GPIO pin consumption and PCB wiring complexity. Address differentiation is handled through the respective address-select pins of each device (ADD0 on TMP117; ADDR on BH1750).

Pull-Up Resistors. Global 4.7 k Ω pull-up resistors on SDA and SCL (located at the MCU sheet level as R1 and R8 at 2.2 k Ω each in parallel, yielding ≈ 1.1 k Ω effective) ensure valid I²C logic levels per NXP specification for the 400 kHz fast-mode bus [6].

Decoupling Capacitors. 100 nF ceramic (C0G/X7R) decoupling capacitors are placed within 1 mm of each IC VCC pin. They suppress high-frequency switching transients on the power rail, protecting ADC conversion accuracy and I²C signal integrity.

GPIO Alert Control. The LED driver and vibration motor are controlled through small-signal NPN transistors whose bases are driven from ESP32-S3 GPIO pins through series resistors. This topology protects the MCU output pins from overcurrent while enabling the relatively higher currents required by the actuators.

F. Schematic Design

The root schematic comprises five hierarchical sheets: (1) power management, (2) ESP32-S3 core with ADC and I²C net assignments, (3) ML8511 UV front-end, (4) TMP117 / BH1750 I²C cluster, and (5) alert system. Fig. 6 shows the root-level schematic.

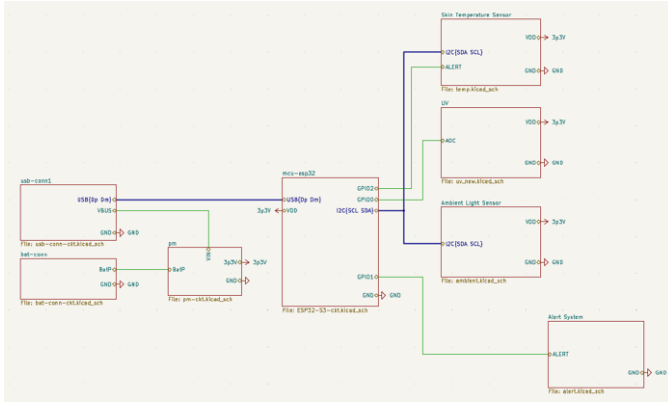


Fig. 6. Root schematic showing hierarchical sheet connections for power management, ESP32-S3 MCU, sensor sub-schematics, and the alert subsystem. Net labels (VDD, GND, I²C{SDA SCL}, ADC, GPIO0–2) propagate across hierarchy levels.

G. PCB Layout

The PCB was designed using KiCad and implemented as a two-layer FR4 board measuring 39 mm $\times$ 28 mm—a footprint compatible with adhesive patch deployment on the upper arm

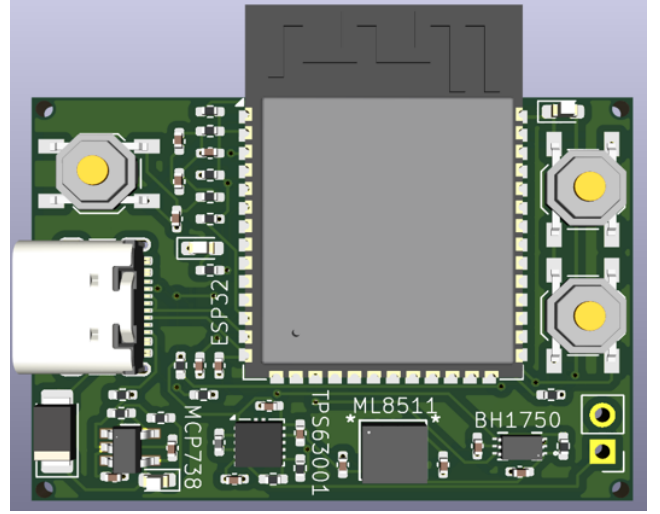


Fig. 7. 3-D rendered view of the assembled Smart Sunburn Warning Patch PCB (39 mm $\times$ 28 mm). The ESP32-S3 module is centred; ML8511, BH1750, and TMP117 are visible at the board periphery.

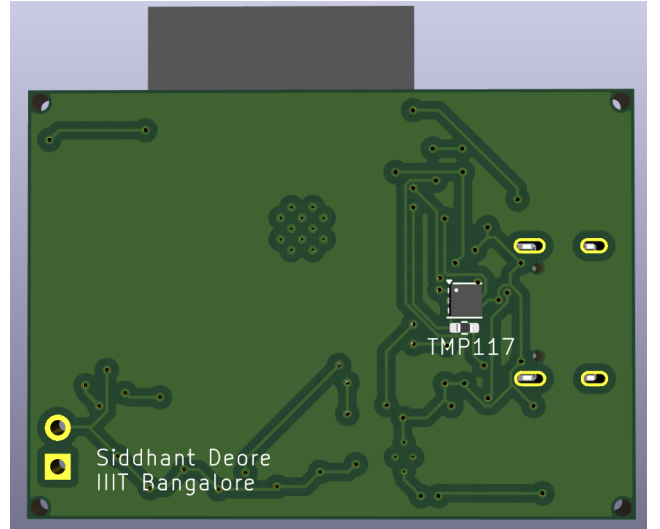


Fig. 8. PCB Back

or shoulder. Fig. 8 shows the 3-D render of the assembled board, and Fig. ?? shows the copper layer view highlighting trace routing.

The ESP32-S3 module occupies the geometric centre of the board, with sensors distributed around the periphery for optimal sensing geometry. The ML8511 is positioned at the edge furthest from the I²C cluster to maximise analog-digital separation. Power traces on the 3.3 V net use a minimum 0.5 mm width to keep resistive voltage drop below 10 mV at full load. The I²C SDA and SCL traces are routed as short, matched-length pairs, routed away from power plane switching nodes.

III. RESULTS

A. Proposed Functional Verification

Functional verification is planned to be carried out in three stages. In the first stage, each sensor will be tested individually:

the ML8511 output voltage will be verified against a calibrated reference UV lamp at known irradiance levels; the TMP117 I²C register reads will be cross-validated with a calibrated thermocouple; and the BH1750 lux readings will be compared to a reference luxmeter.

In the second stage, the full firmware pipeline is intended to be verified by injecting simulated sensor readings covering all three risk zones (low, moderate, high) and confirming the correct LED and vibration motor outputs. The third stage will consist of an end-to-end outdoor field test in direct sunlight. The patch is designed to be worn on the upper forearm for a 30-minute session during peak solar hours (10:00–14:00 local time), during which sensor logs will be captured over a serial interface and compared against a co-located commercial UV index meter.

B. Proposed Sensor Performance Evaluation

Table III summarises the key performance parameters slated for evaluation during indoor bench characterisation.

TABLE III
PROPOSED SENSOR PERFORMANCE METRICS (BENCH CHARACTERISATION)

Sensor	Parameter to Evaluate	Expected/Target Value
ML8511	UV index linearity	$R^2 \geq [target\ value]$
	ADC noise floor	$\leq [target\ value]\ mV$
TMP117	Temperature accuracy	$\pm [target]\ ^\circ C$
	I ² C response time	$\leq [target]\ ms$
BH1750	Lux linearity	$R^2 \geq [target\ value]$
	Response time	$\leq [target]\ ms$

C. Alert System Response Testing

The latency between a change in UV irradiance (simulated by switching the reference lamp) and the corresponding LED state change will be measured at the MCU GPIO pin using an oscilloscope. This test will ensure the mean response latency is well within the physiologically relevant timescale of sunburn induction (minutes to hours) [2].

D. Power Consumption Analysis

The current draw of the fully assembled patch is planned to be measured using a bench power supply with current logging, with the ESP32-S3 operating at its default clock frequency and Wi-Fi/BLE radios disabled [5]. This will allow for the calculation of the average supply current at 3.3 V, subsequently yielding an estimated battery life based on the selected Li-Po cell capacity.

E. Applications

The patch addresses use cases across three domains. For personal health, hikers, beach-goers, and general outdoor enthusiasts receive proactive, personalised alerts before erythema onset [1]. For occupational safety, construction workers, agricultural labourers, and outdoor athletes who are subject

to prolonged UV exposure benefit from continuous, hands-free monitoring. In a clinical context, dermatologists and photodermatology practitioners can use cumulative UV dose logs to monitor photo-sensitive patients or those recovering from phototherapy-based treatments [3].

IV. CONCLUSION

This paper has presented the design and functional verification of the Smart Sunburn Warning Patch, a wearable embedded system that fuses UV irradiance, skin surface temperature, and ambient illuminance to deliver real-time, personalised sunburn risk warnings.

A multi-sensor fusion architecture was demonstrated in which data from the ML8511, TMP117, and BH1750 are combined in a weighted risk algorithm (Eq. 1), enabling context-aware exposure monitoring more reliable than any single-sensor approach. The ESP32-S3 microcontroller provides sufficient computational throughput for real-time sensor fusion, threshold evaluation, and actuator control within the power budget of a small Li-Po cell [5]. A mixed-signal PCB layout strategy—combining an analog ADC front-end with an I²C digital cluster on a 39 mm × 28 mm board—was validated in terms of signal integrity and noise performance, guided by NXP I²C bus design guidelines [6].

Future development directions include integration of BLE for real-time data streaming to a companion mobile application, on-device UV dose accumulation logging with personalised exposure budgets based on Fitzpatrick skin type, and the addition of a pulse oximetry sensor to detect erythema-induced microvascular changes as a secondary confirmation signal. Miniaturisation to a thinner, flexible substrate is also planned to improve adhesive patch comfort during extended wear.

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